

Alexander Belik,

Theoretical Physics Graduate



About Me

Fourth-year Theoretical Physics student at Trinity College Dublin with strong interests in particle physics and modern theory. Experienced in research, programming, and data analysis, including experimental work and EdTech development. Motivated to connect theory with practical applications, with drive for innovation.



Work experience

present
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Oct 2025

Teaching Assistant

Trinity College Dublin, Dublin, Ireland

- Lead weekly tutorials and support problem-solving sessions aligned with lectures.
- Grade assignments and exams, providing clear and constructive feedback.

Dec 2025
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Jun 2024

Software Engineer (EdTech)

Grinds360, Dublin, Ireland

- Built Python automation tools to streamline content and workflow pipelines for videos and interactive lessons.
- Contributed to product development, including UX iteration in Figma and feature implementation.

Sep 2025
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Jun 2025

Research Assistant (CRANN)

Trinity College Dublin (CRANN Institute), Dublin, Ireland

- Designed and implemented Echo State Network (ESN) reservoir computing models to learn and predict the time evolution of quantum systems from simulation data.
- Systematically analysed model stability, convergence, and predictive accuracy across a range of reservoir hyperparameters.

Sep 2024
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May 2024

Research Assistant (Experimental / Computational)

Trinity College Dublin, Dublin, Ireland

- Created a control system for spectroscopy measurements on ferrofluids and liquid crystals under different experimental conditions.
- Built Python tools for automated data acquisition and experiment control.

Jan 2024
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Jul 2021

Exam Corrector

The Dublin Academy of Education, Dublin, Ireland



Projects / Research highlights

2025

Echo State Networks for Quantum Dynamics

Reservoir computing models for learning quantum time evolution; evaluated performance against numerical baselines.

2024

Automated Spectroscopy Data Pipeline

Python-based tooling for automated experiment control and structured data logging. <https://arxiv.org/abs/2504.19633>



Contact

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Location

Dublin, Ireland



Links

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github.com/Abelik1



Education

B.A. Theoretical Physics 2022–2026 (expected)

Trinity College Dublin, Dublin, Ireland
Focus: quantum mechanics, statistical mechanics, QFT, computational physics.

Leaving Certificate 2020–2022
The Dublin Academy of Education, Dublin, Ireland



Skills

- Python (NumPy, automation, tooling)
- PyTorch / machine learning
- C/C++ Computer Simulations
- Scientific computing & data analysis
- LaTeX (technical writing)
- Git / GitHub
- UX collaboration (Figma)



Languages

English

Fluent

Russian

Intermediate

Spanish

Basic



Interests

Programming projects, artificial intelligence, travel, snowboarding, and speed-solving Rubik's Cubes (sub-25s).